

Skill week-6

main.py code:

```
from fastapi import FastAPI, HTTPException, Depends
from fastapi.security import (
    OAuth2PasswordBearer,
    OAuth2PasswordRequestForm )

from jose import jwt
app = FastAPI()
SECRET_KEY = "my-secret-key"
ALGORITHM = "HS256"
oauth2_scheme = OAuth2PasswordBearer(
    tokenUrl="login"
)

@app.post("/login")
def login(
    form_data: OAuth2PasswordRequestForm = Depends()
):
    if (
        form_data.username == " Chaitanya"
        and form_data.password == "chaitanya123"
    ):
        payload = {
            "sub": form_data.username,
            "role": "student"
        }
        token = jwt.encode(
            payload,
            SECRET_KEY,
            algorithm=ALGORITHM
```

```

    )
    return {
        "access_token": token,
        "token_type": "bearer"
    }
    raise HTTPException(
        status_code=401,
        detail="Invalid username or password"
    )

def get_current_user(
    token: str = Depends(oauth2_scheme)
):
    try:
        payload = jwt.decode(
            token,
            SECRET_KEY,
            algorithms=[ALGORITHM]
        )
        return payload
    except Exception:
        raise
        HTTPException(
            status_code=401,
            detail="Invalid or expired token"
        )

@app.get("/profile")
def profile(
    user=Depends(get_current_user)
):
    return {
        "message": "Welcome to your profile",

```

```
"user_id": user["sub"],  
"role": user["role"]  
}
```

POST

POST /login Login

Parameters

No parameters

Request body required

grant_type

string | (string | null)
pattern: ^password\$

password

☐ Send empty value

username * required

string

Chaitanya

password * required

string(\$password)

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

string

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Execute

Response:

Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/login' \  
  -H 'accept: application/json' \  
  -H 'Content-type: application/x-www-form-urlencoded' \  
  -d 'grant_type=password&username=Charan123&password=priyanka123&scope=&client_id=string&client_secret=*****'
```

Request URL

```
http://127.0.0.1:8000/login
```

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiQcm15YW5rYSIsInJvbGUiOiI3ZdWRvZ6C0In0.5gVUoqQUVvJo4VbvX_x_-7k2c1D0fBm8qk1DnJDDknc", "token_type": "bearer" }</pre> Response headers <pre>content-length: 169 content-type: application/json date: Tue, 15 Sep 2026 05:35:18 GMT server: uvicorn</pre>

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Authorization:

Available authorizations x

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.
API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: login
Flow: password
username: Chaitanya
password: *****
Client credentials location: basic
client_secret: *****

[Logout](#) [Close](#)

GET:

GET /profile Profile 🔒 ^

Parameters Cancel

No parameters

[Execute](#) [Clear](#)

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC79.eyJzdWIiOiJ0xc15W5Y5IsInJvbGUiOiJzdWI6d9lmdj.eyJ0c0QwJ3J4VvbnVX_z_-7kZc1D0F8Bqk1Dn300knc'
```

Request URL

```
http://127.0.0.1:8000/profile
```

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to your profile", "user_id": "Chaitanya", "role": "student" }</pre>📄 Download Response headers <pre>content-length: 75 content-type: application/json date: Tue, 15 Sep 2026 05:40:14 GMT server: uvicorn</pre>